

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – INCIDENT RESPONSE (MONITORING & MITIGATION)

Course Code:	CSA5803	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO5 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	Develop Incident Response Plan, Analyze Security Monitoring Dashboard, Recommend Mitigation Strategies and Technical Justification.		
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Section / Batch:	Slot C	Date:	27/08/26

Instructions to Students

- Develop an incident response plan based on the given security incident scenario and identified findings.
- Analyze the provided security monitoring dashboard, alerts, logs and indicators to determine incident status and affected assets.
- Recommend prioritized mitigation strategies that directly address the identified threats, vulnerabilities and incident impact.
- Provide clear technical justification for each major response and mitigation decision using evidence from the scenario.
- Present the response plan, dashboard analysis and recommendations clearly and professionally within the prescribed submission format.

Question Type	Focus Area	Questions	Marks
INCIDENT RESPONSE	Monitoring & Mitigation	Q1	Q1 – 100
GRAND TOTAL:		100 Marks	

Incident Response Plan, Security Monitoring Dashboard Analysis, Mitigation Strategies and Technical Justification

1. Incident Response Plan

The incident involves a **student device scanning multiple internal VLANs** in a campus network. This activity may indicate malware infection, unauthorized scanning, or an attempt to discover vulnerable systems.

Step 1: Detection

- IDS/IPS detects abnormal scanning activity.
- Security dashboard generates an alert.
- Record the source IP address, MAC address, username and timestamp.

Step 2: Identification

- Identify the student device and its connected switch port/Wi-Fi access point.
- Determine the VLANs and systems targeted by the scan.
- Check DHCP, RADIUS, firewall and IDS/IPS logs.

Step 3: Containment

- Immediately isolate the suspicious device using **NAC**.
- Move the device to a quarantine/remediation VLAN.
- Block unnecessary inter-VLAN traffic using firewall rules and ACLs.

Step 4: Investigation

- Analyze the source and destination IP addresses.
- Check scanned ports and protocols.
- Determine whether the activity was caused by malware, unauthorized software, compromised credentials or legitimate testing.
- Preserve logs for forensic investigation.

Step 5: Eradication

- Remove malware or unauthorized scanning tools.
- Patch the operating system and applications.
- Reset compromised passwords or credentials.
- Correct network or security misconfigurations.

Step 6: Recovery

- Perform endpoint security and vulnerability checks.
- Reconnect the device only after it satisfies security requirements.
- Monitor the device for further suspicious activity.

Step 7: Documentation

- Record the incident, root cause, affected systems and actions taken.
- Update security policies and network controls to prevent recurrence.

2. Analysis of Security Monitoring Dashboard

The dashboard should be examined for indicators of abnormal behavior.

Important indicators

Source IP/MAC:

Identifies the student device generating the traffic.

Destination IPs:

Multiple destinations across different VLANs indicate possible network reconnaissance.

Ports:

Scanning common ports such as SSH, HTTP, HTTPS, RDP or database ports may indicate an attempt to identify available services.

Traffic volume:

A sudden increase in connection attempts is an important warning sign.

VLAN access:

A student device attempting to access faculty, server or administration VLANs is highly suspicious.

IDS/IPS alerts:

Repeated reconnaissance or exploit-related alerts increase the severity of the incident.

Dashboard conclusion

The combination of **one student endpoint + multiple internal VLANs + high connection attempts + unusual ports** indicates potential reconnaissance or compromise. The device should therefore be investigated and temporarily isolated.

3. Recommended Mitigation Strategies

A. Network Access Control (NAC)

NAC should be used to authenticate and evaluate every device before allowing network access.

Benefits:

- Identifies users and devices.
- Enforces security policies.
- Detects non-compliant endpoints.
- Automatically quarantines suspicious devices.
- Prevents unauthorized devices from accessing internal networks.

Technical justification:

NAC provides **identity-based access control** and supports the principle of least privilege.

B. VLAN Isolation

Campus networks should be separated into different VLANs:

- Student VLAN
- Faculty VLAN
- Laboratory VLAN
- Server VLAN
- Administration VLAN
- Guest Wi-Fi VLAN

Communication between VLANs should pass through controlled firewalls or Layer-3 ACLs.

Technical justification:

VLAN segmentation reduces the **attack surface and lateral movement**. Even if a student device is compromised, the attacker cannot freely access critical systems.

C. IDS/IPS

An **Intrusion Detection and Prevention System** should continuously monitor traffic.

- IDS detects and alerts on suspicious activity.
- IPS can automatically block malicious traffic.
- Detect port scanning and reconnaissance.
- Identify known attack signatures.
- Generate alerts for security administrators.

Technical justification:

IDS/IPS provides real-time visibility into network behavior and helps detect attacks that bypass normal access controls.

D. Firewall and ACLs

Implement restrictive rules between VLANs.

For example:

Student VLAN → Server VLAN: Allow only required services.

Student VLAN → Administration VLAN: Deny by default.

Guest VLAN → Internal VLANs: Deny.

Faculty VLAN → Required academic services: Allow.

Technical justification:

Firewall and ACL rules enforce **least-privilege communication** and prevent unauthorized inter-VLAN access.

E. Endpoint Remediation

The suspicious device should be quarantined and inspected.

- Run EDR/antimalware scans.
- Check running processes and network connections.
- Remove unauthorized software.
- Install security patches.
- Reset compromised credentials.
- Reconnect only after the device passes security checks.

4. Technical Justification

A **defense-in-depth architecture** should be implemented:

NAC → VLAN Isolation → Firewall/ACL → IDS/IPS → Endpoint Security → Centralized Logging

Each layer provides a different security function.

- **NAC:** Controls device and user access.
- **VLAN:** Provides network segmentation.
- **Firewall/ACL:** Controls permitted communication.
- **IDS/IPS:** Detects and blocks suspicious traffic.
- **Endpoint Security:** Detects and removes malware.
- **Centralized Logging:** Supports investigation and incident response.

Conclusion

The suspicious student device should first be **detected and identified**, then **isolated using NAC**, investigated through the security monitoring dashboard and logs, and finally **remediated before being allowed back onto the network**. VLAN segmentation, firewall/ACL controls and IDS/IPS provide multiple layers of protection and significantly reduce the risk of lateral movement and unauthorized access to critical campus systems.